

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

CM2A - Economic Modelling

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) Two portfolios of assets, A and B having the following expected returns:

State	Portfolio A	Portfolio B	Probability
Good	18%	4%	0.37
Bad	6%	9%	0.22
Average	12%	7%	0.41

The S.D of returns for Portfolio A and Portfolio B are:

- a) 12.90%, 6.33%
- b) 0.204%, 0.038%
- c) 4.52%, 1.939%
- d) 13.669%, 6.62%
- e) 1.868%, 0.438%

[2 marks]

Q.2) A future contract is on a stock, currently trading at Rs 200 and a Rs.4 dividend payable 3 months from now, what should be the fair price as set today for the 6-month future contract when the rate of interest in the market is 4%?

- a) 200
- b) 200.42
- c) 204.04
- d) 203.06
- e) 199.50

[2 marks]

Q.3) A company has issued a 10-year debt with a nominal face value of 10 cr. The overall asset value of the company is 40 cr. Assuming we use the Merton model to value the debt issued by the company. Say the total asset value of the company after 10 years is denoted by $F(10)$, payoff to the bond holder would be as follows:

- a) $\text{Max}[10, F(10)]$
- b) $\text{Min}[10, F(10)]$
- c) $10 - \text{Min}[40 - F(10), 0]$
- d) $10 - \text{Max}[0, F(10) - 10]$
- e) $10 - \text{Max}[40 - F(10), 0]$

[2 marks]

Q.4) An investor has decided to model the share price of ABC using the continuous-time lognormal model. Using historical data, the investor has estimated the annual drift and volatility parameters to be 7% and 22%. The current share price for ABC is Rs. 1150.

The mean and SD of ABC share price in exact 2 years will be:

- a) 1387, 515
- b) 1391, 347
- c) 1398, 477
- d) 1492, 349
- e) 1575, 429

[2 marks]

Q.5) Claims occur on a portfolio of insurance policies according to a Poisson process at a rate λ . The mean claim amount is μ . The insurance company calculates premiums using a loading of θ and has an initial surplus of U ($< \mu$). Premiums are received continuously. The probability of ruin of first claim is :

- a) $\exp(-\lambda/(1+\theta)*(1-U/\mu))$
- b) $1-\exp(-\lambda/(1+\theta)*(1-U/\mu))$
- c) $\exp(-\mu/(1+\theta)*(1-U/\mu))$
- d) $1-\exp(-\theta/(1+\theta)*(1-U/\mu))$
- e) $\exp(-\lambda/\theta*(1-U/\mu))$

[2 marks]

Q.6) In an Efficient Stock Market, the following statements are made about the stocks?

- a. The price of every security fully reflects all available information
- b. The price of every security is their TRUE Value
- c. There is no information asymmetry

I. a is Correct

II. a and b are Correct

III. a and c are Correct

IV. b and c are correct

V. All are correct

[2 marks]

Q.7) A portfolio manager is building a portfolio with forward contracts which pays twice of share price at maturity. She is trying to vega hedge the portfolio. If the portfolio is vega hedged then

- a. The portfolio will be insensitive to changes in volatility of the underlying share
- b. The portfolio will be insensitive to changes in interest rates
- c. Could not estimate volatility of the underlying shares accurately
- d. The portfolio will be sensitive to changes in volatility of the underlying share

I. only d

II. only b

III. both b and d

IV. both a and c

V. None of the above

[2 marks]

Q.8) The investor is having a portfolio of the following two options

- Bought an European call option for Rs. 2 with Strike price of Rs. 50 and the price of underlying share at expiry is Rs. 51
- Sold an European put option for Rs. 3 with Strike price of Rs. 55 and the price of underlying share at expiry is Rs. 52

The profit or loss for the investor of his portfolio at expiry is

I. Loss of Rs. 1

II. Profit of Rs. 2

III. Loss of Rs. 8

IV. Loss of Rs. 3

V. Loss of Rs. 2

[2 marks]

Q.9) An insurance company writes a Fasal Bima Yojana policy for the farmers. At the end of each year, the farmer's crop is assessed, and if it is less than 200 tonnes, it is deemed to have failed. If the crop fails for two years in a row the insurance policy pays out Rs. 20 Lakhs and then is immediately terminated. Premiums are Rs. 45000 per month, paid in advance, and there are no expenses. This is the only policy the insurance company writes, and it has initial surplus $U > 0$.

Denote by $\Psi(U,t)$ the probability of ruin by time t , measured in years; and $\Psi(U)$ as the ultimate probability of ruin.

Below mentioned are 4 conditions:

a) $\Psi(U,1) < \Psi(U,1.5)$

b) $\Psi(U,2.5) = \Psi(U,3.33)$

c) $\Psi(U,3) < \Psi(U,4)$

d) $\Psi(U,4) = \Psi(U)$

i) a and b are true

ii) Only d is true

iii) b and c are true

iv) a, c and d are false

v) a and d is true

[2 marks]

Q.10) For a company with x years of debt with a nominal face value of F and overall assets worth A , the pay-off for the equity holders of a company can be regarded as having:

- a) European call option with maturity of x years and strike price of F
- b) European call option with maturity of x years and strike price of $(A-F)$
- c) European put option with maturity of x years and strike price of $(A-F)$
- d) European put option with maturity of x years and strike price of F
- e) American call option with maturity of x years and strike price of F

[2 marks]

Q.11) Which of the following provide evidence against the weak form of the EMT?

I. Changes in stock prices are somewhat predictable from past data.

II. Changes in stock prices are random and unpredictable.

III. Insiders of company can consistently profit from insider trading.

a) I and II only

b) I and III only

c) I only

d) I, II and III

e) None of the above

[2 marks]

Q.12) Which of the following behaviours of insured policyholders as 'Adverse selection'?

I. A person with co-morbid conditions availing a group term insurance through the employer

II. Healthy life not paying the renewal premium for a term insurance in the last 2 years

- III. A vehicle owner parking the car below a tree during windy rainy season
 IV. A biker riding the bike carelessly after buying additional accidental cover

- i) Option I and II
 ii) Option I and III
 iii) Option III and IV
 iv) Option II and IV
 v) None of the above

[2 marks]

Q.13) Describe informational efficiency in the context of the Efficient Markets Hypothesis. List examples of effects that have been claimed to exist in stock markets that might be considered examples of informational inefficiency. [4 marks]

Q.14) The stock exchange in the country of XYZ is flourishing with a liquid derivatives market. The Stock Exchange maintains its benchmark index called ABC, with futures and European options traded for this index. Following is the details of option prices expiring 3 months from now. Find the missing entry.

Strike	Call	Put
1,000	199.08	31.73
1,100		65.21
1,200	85.68	114.86

Calculate the fair price of the 6 months futures contract entered today on ABC.

[5 marks]

Q.15) A portfolio consists of two assets A and B with the correlation co-efficient between their returns is 0.5. An investor has invested an equal amount of ₹1000 in both of these assets. Given that the daily volatilities of both the assets are 2%, find the 10-day 99% VaR for the Portfolio. [5 marks]

Q.16) Neeraj Chopra is an Indian athlete who will be competing at the Olympic Games. He has taken out special insurance which pays out if he is injured. If the injury is so serious that his career has ended, the policy will pay 100 Crores and will be terminated. If he is injured but recovers, the insurance payment will be Rs 10 Crore and the policy will continue. The insurance company's underwriters believe that the probability of an injury in any year is 0.18, and that the probability of more than one injury in a year can be ignored. If he is injured, there is a 85% chance that he will recover. Annual premiums are paid in advance, and the insurance company pays claims at the end of the year. Assume that this is the only policy that the insurance company writes and that it has an initial surplus of Rs. 10 Crore.

Calculate the annual premium charged assuming the insurance company uses a premium loading of 25%. Also determine $\psi(10\text{Crore}, 2)$. [7 marks]

Q.17)

- i) List the assumptions, additional to those used in the modern portfolio theory, of the CAPM. [2]
 ii) Prove that market has a beta of 1. [2]
 iii) In the market mentioned in ii, the following details of 2 asset classes is mentioned. Calculate the beta of each of the securities.

Rate of return per annum							
State	Probability	Asset 1	Asset 2		Variance/Covariance		
1	0.1	6%	10%			Asset 1	Asset 2
2	0.2	11%	14%		Asset 1	0.000585	0.000925

3	0.3	9%	11%		Asset 2	0.000925	0.001621
4	0.4	5%	4%				
Market Cap		5,000.00	15,000.00				

[2]

- iv) Determine the value of the risk free rate of interest which is consistent with the result obtained in (iii), assuming CAPM holds. [2]

[8 marks]

Q.18) A trader wants to determine the value of 6-months European put option on a non dividend paying stock using a two-step binomial tree. The current price of the share is ₹475 and the payoff is $\text{Max}(450 - S_T, 0)$ where S_T is the price of the share in 6 months. The investor assumes that for the first period (first 3 months) the share price will increase by 12% or decrease by 6% and the continuously compounding risk-free rate of interest is 3% (per 3 months). For the next 3 months the expected risk-free rate is assumed to be 2% (per 3 months) and the share price to increase by 15% or decrease by 7%.

Calculate the value of the European put option using the 2 step binomial tree.

Explain what is meant by recombining binomial tree. State the main advantage of using it in modelling the share price movements. [9 marks]

Q.19) The table below gives the cumulative incurred claims by year and premiums received for a third party claim of 4 wheeler policy (Figures in thousands). Claims paid till date total 2 crores. The ultimate loss ratio is expected to be in line with the 2022 accident year.

Accident Year	Development Year				Earned Premium
	0	1	2	3	
2022	4000	4500	5125	5300	5400
2023	4400	4900	5500		5900
2024	4500	5150			6100
2025	5000				6250

Use the Bornhuetter-Ferguson method to calculate the total reserve required to meet the unpaid claims, assuming that the claims are fully developed by the end of development year 3. (Ignore inflation.)

[9 marks]

Q.20) Consider a zero-coupon bond B_t with three years to maturity. The bond pays Rs.100 at maturity if it has not defaulted, or Rs. 20 if it has defaulted. The continuously compounded risk-free rate is r . In a two-state model the default intensity is λ under the probability measure P , and the bond price is:

$B_t = 20 \exp(-r(2-t))$ if the bond has already defaulted by time t , or

$B_t = \exp(-r(2-t)) [20(1 - \exp(-\lambda(2-t))) + 100 \exp(-\lambda(2-t))]$ if the bond has not yet defaulted by time t .

- i) Show that P is an equivalent martingale measure. [4]

A derivative pays Rs. 25 at time 2 if the bond has defaulted and Rs. 0 otherwise.

ii)

- a) Determine a constant portfolio containing the bond and cash which replicates this derivative. [3]

- b) Derive an expression for the arbitrage-free price for the derivative at time 0 in terms of r and λ .

[2]

- iii) Explain how your answers to parts (i) and (ii) are related through the value of the portfolio in part (ii) also being a martingale. [3]

[12 marks]

Q.21) The government has introduced a new 20 year bond which is trading at par with coupon rate of 7% per annum paying coupon annually with a market yield of 7%. The risk free rate is assumed to be 7% as well.

- i) Calculate the price of an at the money call option where the volatility is 7% and the option is for a tenure of 2 years. [2]
- ii) An investor is looking to replicate the same using cash and the underlying. Provide the replicating strategy for the same. [3]
- iii) A year later, yields have increased to 7.5% rest all conditions remaining the same. Calculate the market price of the bond. [2]
- iv) Calculate the price of the call option now when time to maturity is 1 year left. [2]
- v) Define the replicating strategy for the call option now. [3]
- vi) Also mention the increase/decrease in the replicating strategy bonds and cash position and comment on the same. [5]

[17 marks]
